UNEMPLOYMENT ANALYSIS USING PYTHON

Project Report

CodeAlpha Internship — Task 2

A data analysis project exploring unemployment patterns across Indian regions, differences between rural and urban areas, and changes during the early COVID-19 period.

Project focus	Unemployment trends, regional comparisons and labour force participation.
Tools used	Python, pandas, NumPy, Matplotlib and Seaborn.
Dataset period	May 2019 to June 2020.

1. About the Project

For this project, I used Python to explore unemployment data for India. I looked at how unemployment rates differed between regions, compared rural and urban observations, and examined the changes recorded during the early COVID-19 period. The aim was to use the data to identify patterns and present the results in a clear way.

2. Project Aim

The aim was to understand the main patterns in the unemployment dataset through data cleaning, exploratory analysis and visualisation. I also wanted to see how unemployment changed over time, whether rural and urban results differed, and how labour force participation compared between the two areas.

3. Understanding the Dataset

The dataset contained 754 rows and 7 columns before cleaning. It included Region, Date, Frequency, Estimated Unemployment Rate (%), Estimated Employed, Estimated Labour Participation Rate (%) and Area. The observations covered May 2019 to June 2020 and included rural and urban records.

4. Preparing and Exploring the Data

I loaded the CSV file into a pandas DataFrame and inspected the first few rows, the column names, data types and summary statistics. I also checked the dataset for missing values and duplicate records.

• I found 14 completely empty rows and removed them from the dataset.

• The cleaned dataset contained 740 rows and 7 columns.

• I checked for missing values and duplicate rows after cleaning; both checks returned zero.

• I converted the Date column into a datetime format so that the records could be grouped by month.

• I grouped the data by Region, month and Area to calculate average unemployment and labour force participation rates.

• I used bar charts and a line chart to make the main comparisons and monthly pattern easier to understand.

5. Overall Unemployment Rate

The average estimated unemployment rate across the observations in the cleaned dataset was approximately **11.79%**. This is the mean of the dataset observations over the period covered, rather than a current unemployment figure or a single official national estimate.

6. Regional Unemployment Results

The regional averages showed clear differences across the regions included in the dataset. Tripura had the highest average unemployment rate, at approximately **28.35%**. Meghalaya had the lowest, at approximately **4.80%**.

Region	Average unemployment rate
Tripura	28.35%
Haryana	26.28%
Jharkhand	20.59%
Bihar	18.92%
Meghalaya	4.80%

7. Monthly Unemployment Trend and COVID-19 Period

The monthly averages were around 9–11% from May 2019 to March 2020. They then increased sharply to approximately **23.64%** in April 2020 and **24.88%** in May 2020, before falling to approximately **11.90%** in June 2020.

The increase occurred during the early COVID-19 period, when economic activity was disrupted. The dataset shows a clear change during this period, although this analysis by itself cannot prove that the pandemic or lockdown measures were the only cause.

Month	Average unemployment rate
February 2020	9.96%
March 2020	10.70%
April 2020	23.64%
May 2020	24.88%
June 2020	11.90%

8. Rural and Urban Unemployment

The average unemployment rate was approximately **10.32%** in rural areas and **13.17%** in urban areas. In this dataset, the urban average was higher by about 2.84 percentage points.

Area	Average unemployment rate
Rural	10.32%
Urban	13.17%

9. Labour Force Participation

The average estimated labour force participation rate was approximately **44.46%** in rural areas and **40.90%** in urban areas. This means the rural average was higher by about 3.56 percentage points in the observations analysed.

Area	Average labour force participation rate
Rural	44.46%
Urban	40.90%

10. What I Learned

Working on this project helped me practise the main steps involved in a data analysis task. I learned how to load a dataset, inspect and clean it, work with dates, calculate averages using groups, and create charts to communicate results. I also gained a better understanding of how to compare different parts of a dataset rather than relying on one overall figure.

11. Conclusion

Overall, this project helped me use Python to explore unemployment patterns in India. The analysis showed differences between regions, higher average unemployment in urban than rural observations, and higher average labour force participation in rural observations. It also showed a sharp increase in unemployment during April and May 2020, followed by a decline in June. These findings describe patterns in the dataset and do not establish cause and effect.

The project gave me practical experience in data cleaning, exploratory data analysis and visualisation. With more recent data and additional information, the analysis could be extended to study how these patterns changed after June 2020.

Tools Used

Python, pandas, NumPy, Matplotlib, Seaborn and Google Colab.

Note: The results in this report are based on the outputs recorded during the completed analysis. The dataset covers May 2019 to June 2020, so the findings should not be treated as current unemployment statistics.